

MACOMB, IL, USA

WMO: 722157

Lat: 40.520N

Lon: 90.652W

Elev: 707

StdP: 14.32

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04949

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.5	2.8	-8.5	3.6	-1.1	-3.9	4.6	4.2	31.0	24.1	27.7	24.0	7.3	300	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	91.5	75.6	89.8	75.1	87.5	73.9	79.6	87.9	77.9	85.8	76.3	83.7	9.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.2	145.5	84.8	75.3	136.3	82.4	73.5	127.9	80.5	43.8	87.9	41.8	85.9	40.1	83.6	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.1	22.9	19.6	DB	-9.6	96.2	6.8	3.5	-14.5	98.7	-18.5	100.8	-22.4	102.8	-27.4	105.4	
			WB	-10.1	82.6	6.7	1.3	-14.9	83.5	-18.8	84.3	-22.6	85.1	-27.5	86.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.0	25.6	28.0	41.1	52.6	64.1	72.3	74.2	72.3	66.7	54.5	41.4	30.2
	DBStd	19.90	12.53	12.35	11.77	9.70	8.84	5.78	5.69	5.62	7.62	9.78	10.24	10.88
	HDD50	2768	758	623	330	80	6	0	0	0	63	293	615	
	HDD65	5746	1222	1037	747	384	129	8	4	8	68	351	710	1078
	CDD50	3510	1	5	53	158	442	670	749	692	502	202	34	2
	CDD65	1015	0	0	4	12	101	229	288	235	120	26	0	0
	CDH74	9773	0	0	38	187	957	2124	2846	2128	1219	271	3	0
	CDH80	3387	0	0	3	34	290	761	1067	714	442	76	0	0
(14) Wind	WSAvg	8.5	10.5	10.5	10.6	11.1	9.5	7.3	5.2	4.7	5.6	8.0	9.8	9.7
(15) Precipitation	PrecAvg	37.7	1.9	1.8	2.5	3.7	4.9	4.3	4.2	3.8	3.2	2.8	2.8	1.9
	PrecMax	51.5	3.3	4.8	5.0	6.3	9.9	9.1	11.3	7.6	5.9	6.6	6.0	3.8
	PrecMin	26.8	0.5	0.4	0.7	1.7	0.8	1.3	1.7	0.9	0.7	1.0	0.3	0.4
	PrecStd	6.2	0.9	1.2	1.3	1.1	2.6	2.2	2.4	1.8	1.6	1.7	1.5	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.9	66.2	79.4	84.4	91.2	93.1	95.6	94.1	93.3	88.0	72.7	61.2
		MCWB	51.1	55.8	64.7	65.0	71.1	76.0	78.4	76.0	71.1	70.6	56.8	57.9
	2%	DB	52.4	57.8	72.0	79.7	86.3	90.4	91.2	90.3	89.3	81.2	66.0	54.9
		MCWB	48.2	50.2	61.1	61.2	68.9	74.7	78.4	76.3	71.8	64.8	56.9	50.7
	5%	DB	46.3	50.8	65.5	74.9	83.0	88.0	88.6	87.4	85.2	75.5	62.4	50.0
		MCWB	42.9	45.7	57.3	60.2	67.1	73.7	76.8	75.3	71.3	63.1	54.8	47.0
	10%	DB	41.0	44.8	59.4	70.4	80.0	84.4	86.0	83.8	81.5	70.8	57.1	45.2
		MCWB	38.2	40.7	52.4	57.4	65.2	71.7	74.9	73.6	67.6	61.2	50.5	42.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.0	58.3	66.4	67.6	74.4	80.1	82.2	80.5	77.3	72.5	62.4	59.0
		MCDB	57.8	64.3	76.3	78.5	85.2	88.5	90.4	88.8	86.1	82.1	66.4	60.8
	2%	WB	48.6	51.7	62.3	64.6	72.2	77.6	80.2	78.7	75.0	69.6	58.8	51.8
		MCDB	51.1	56.9	69.2	74.7	82.5	86.8	88.4	86.8	83.4	77.2	63.9	53.2
	5%	WB	42.8	45.7	58.6	62.4	69.9	75.9	78.4	77.0	73.2	66.6	55.4	46.6
		MCDB	46.6	50.3	65.0	71.2	79.0	84.4	86.1	84.2	81.3	71.6	60.9	49.0
	10%	WB	38.2	40.5	53.2	59.3	67.8	74.0	76.8	75.1	71.2	63.0	51.6	42.2
		MCDB	40.9	44.2	58.2	68.1	76.5	81.8	84.0	81.5	78.1	69.1	56.8	45.2
(35) Mean Daily Temperature Range	5% DB	MDBR	15.7	17.0	18.7	21.9	20.8	19.7	19.6	20.4	23.7	21.7	18.8	14.8
		MCDBR	21.0	26.4	25.1	29.0	24.7	22.7	21.0	22.9	25.8	27.2	25.5	20.6
	5% WB	MCWBR	17.0	18.8	15.0	14.6	10.4	10.5	10.3	11.3	10.9	13.1	16.6	17.0
		MCWBR	20.3	25.5	23.5	23.5	21.3	19.8	19.1	19.0	21.2	20.7	22.6	19.6
(40) Clear-Sky Solar Irradiance	taub		0.291	0.312	0.340	0.386	0.420	0.428	0.444	0.432	0.395	0.353	0.304	0.291
		taud	2.456	2.386	2.402	2.272	2.241	2.271	2.199	2.244	2.310	2.427	2.509	2.467
	Ebn at Noon		278	286	290	281	273	270	264	264	267	268	271	268
		Edh at Noon	25	30	34	41	43	42	45	42	37	29	23	23
(44) All-Sky Solar Radiation	RadAvg		577	883	1182	1509	1758	1974	1985	1739	1492	1002	669	491
	RadStd		68	79	77	123	127	170	114	103	107	111	55	65

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108

Nomenclature: See separate page